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Moonshot: Public R&D and Growth[†]

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We estimate the long-term effect of public R&D on growth in manufacturing by analyzing new data from the Cold War–era space race. We develop a novel empirical strategy that leverages US-Soviet rivalry in space technology to isolate windfall R&D spending. Our results demonstrate that public R&D conducted by NASA contractors increased manufacturing value added, employment, and capital accumulation in space-related sectors. While migration responses were important, they were not sufficient to generate a wedge between local and national effects. The iconic moonshot R&D program had only modest economic effects for both the local and national space-related sectors. (*JEL* E22, H54, L60, N12, N42, N62, O47)

Technological progress plays a central role in theories of economic growth (Solow 1957; Romer 1990; Aghion and Howitt 1992). Because social returns to research and development may be larger than private returns, firms may underinvest in innovation, thus reducing the rate of technological progress (Arrow 1962; Griliches 1992; Bloom, Schankerman, and Van Reenen 2013). Can government-funded R&D fill this gap and generate long-term growth? Despite the fact that governments expend significant resources on R&D every year—over $158 billion within OECD countries in 2020 (OECD 2022)—the answer remains unclear.

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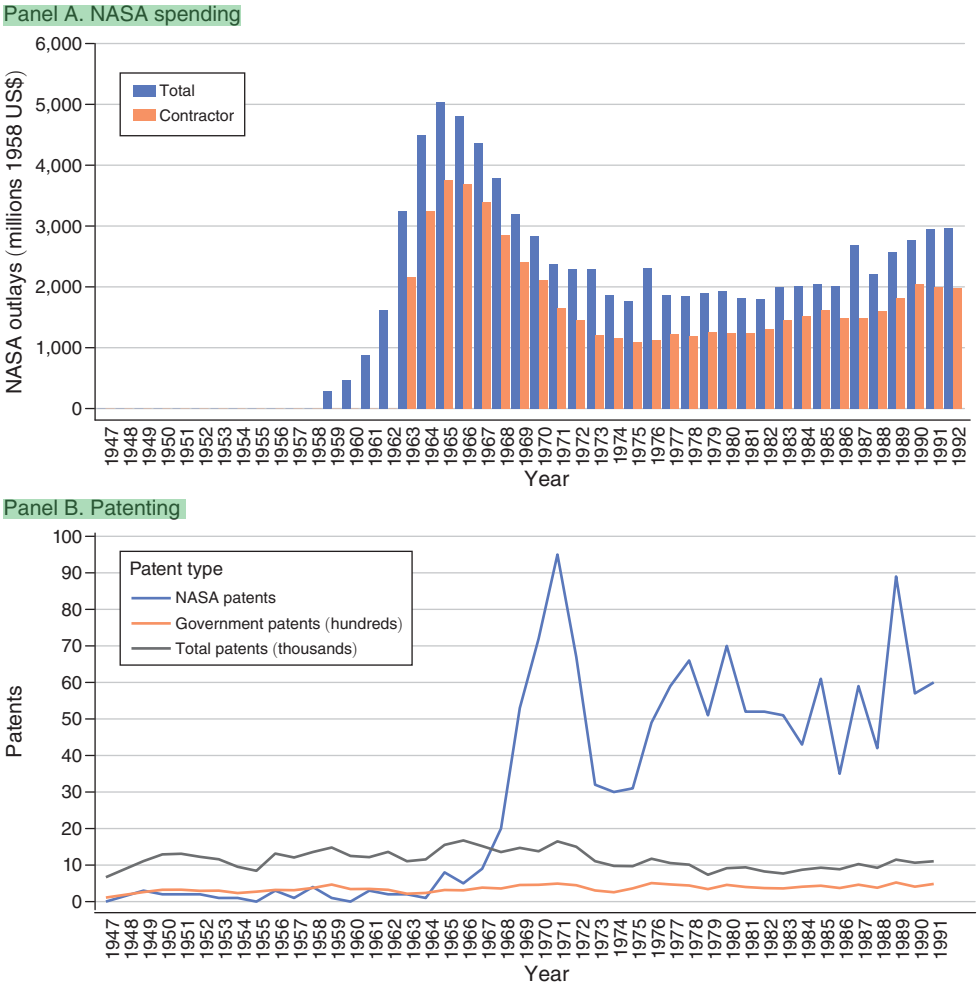


FIGURE 1. NASA SPENDING AND PATENTING, 1947–1992

Notes: NASA spending data are drawn from Van Nimmen and Bruno (1976), Gawdiak (1994), and Rumerman (2000, 2012). NASA spending is measured in US$(1958). See Supplemental Appendix Section 1.4 for the multiple patent data sources. NASA patents include patents assigned to or funded by NASA.

In this paper we provide new evidence on the effects of public R&D on long-term economic growth by analyzing a unique episode in US history: the race to beat the Soviet Union to the moon during the 1960s. The shock of the Soviet launch of the first satellite, Sputnik, in 1957 led to a geopolitical crisis that initiated the creation of NASA in 1958 and launched the race to the moon in 1961. Figure 1 shows that the ambitious mission to send (and return) a manned crew to (and from) the moon led to a massive expansion of federal investment in R&D—NASA received over 0.7 percent of GDP in the mid-1960s (Weinzierl 2018) and employed over 400,000 workers at the peak of the space race. To the extent that we hope to uncover the nuances of how public R&D seeds economic growth, the Cold War–era space race provides a unique episode in modern US economic history to examine in depth.

We analyze the effects of this large R&D windfall on growth in manufacturing in the short- and long-term. Focusing on manufacturing growth is likely to capture the indirect effects of space R&D well because getting to the moon not only required new ideas and technologies, but also the production of real products. Innovations of the space race era were embodied, for example, in spacecraft, satellites, thrusters, navigation and communications equipment, computer software and hardware, and launch infrastructure.

To estimate our models we develop a novel empirical approach to isolate the exogenous variation in NASA contractor R&D. The imperative to win the space race meant that NASA was compelled to rapidly allocate funding to space sector firms already specialized in the technological building blocks needed to complete the mission. NASA did not invest in technologies randomly but sought to harvest any promising space technologies that American firms could supply to win the race to the moon.

We address technology harvesting in two steps. We first utilize the CIA's declassified national intelligence estimates of Soviet space technology from the post-Sputnik era to define the set of technologies demanded by the space mission. We then search for these technologies in US patents before 1958 to determine which US industries in which counties specialized in space-relevant technologies before the space race began. We term county-industries as having "high space capability" if their pre-1958 technological specialization matched post-1958 space technology demand, as seen through the perspective of the Soviet space program and not NASA's. Isolating variation in NASA R&D that is virtually independent of location-specific unobservables, our research design compares changes in outcomes between space industries to other industries, before and after the space race, in county-industries with varying pre-1958 space technology capabilities.



To carry out our empirical analysis we construct a new panel dataset containing highly granular data on US manufacturing and NASA activity for large urban counties from 1947 to 1992. For each industry-county we have digitized the amount that NASA contractors received. We then match this information to manufacturing value added, employment, and labor income from the census of manufactures at the county $\times$ two-digit industry level to estimate our models. We also utilize newly available data on government ownership and funding of patents from Fleming et al. (2019).

NASA spending was highly concentrated in a few sectors (Figure 2, panel A) that grew faster than others in terms of output, employment, capital, and total factor productivity (TFP) over the 1958–1992 period (Figure 2, panel B). These trends may not reflect a causal relationship, however. Our analysis that addresses potential endogenenity of NASA's spending decisions reveals five main results. First, we establish that the space race caused NASA contracting activity to expand more in the industry-county pairs that had already specialized in the building blocks of space technology before Sputnik. The amount of NASA spending and NASA patents expanded significantly relative to other industry-counties that were not already specialized in the rudiments of space technology.

Second, we show that the space race caused manufacturing value added, employment, and capital to expand more in those industry-county pairs that had already specialized in early space technology before Sputnik. One possible concern is that

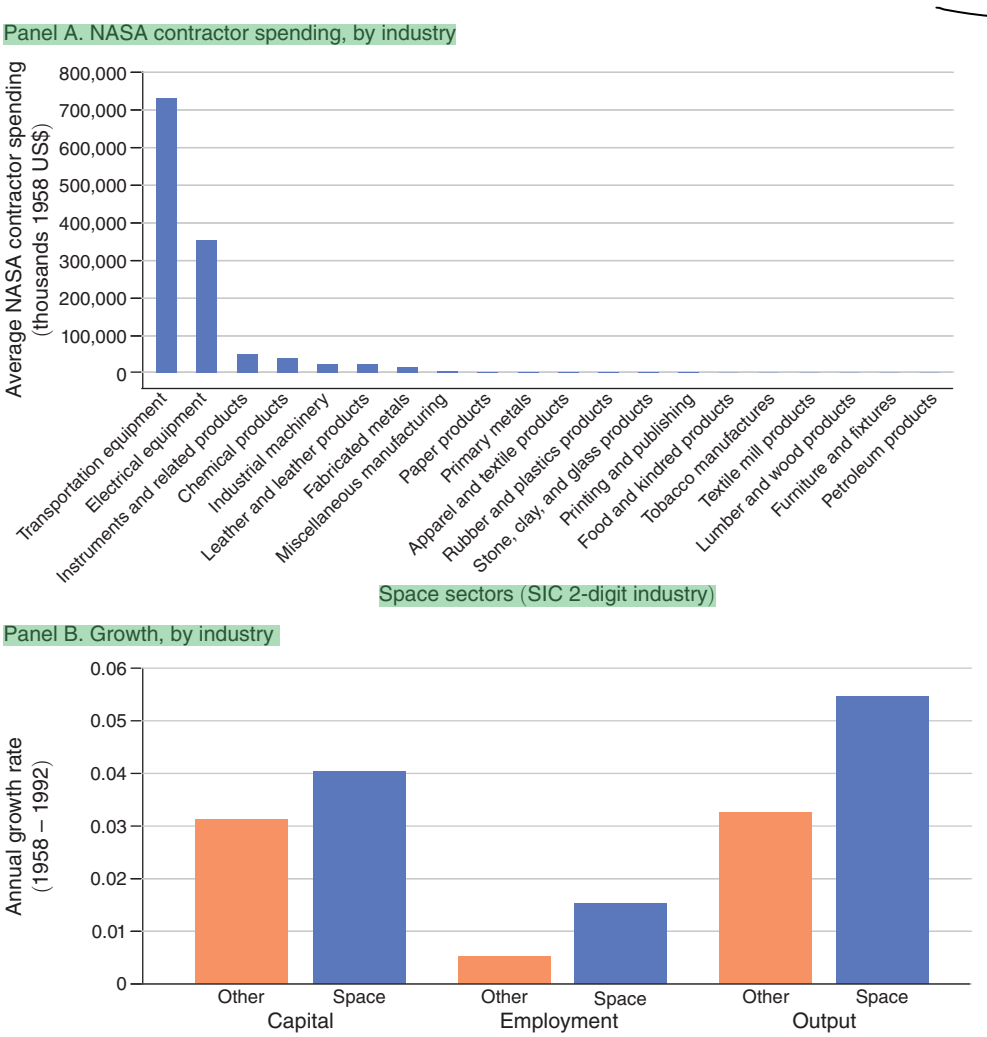


FIGURE 2. NASA CONTRACTOR SPENDING AND GROWTH BY INDUSTRY, 1958–1992

Notes: Data are drawn from Van Nimmen and Bruno (1976); Gawdiak (1994); Rumerman (2000, 2012); and Becker, Gray, and Marvakov (2021).

NASA activity followed trends in manufacturing. We show that there were negligible differential trends before the space race began between high space capability industry-county pairs relative to their counterparts around the country, thus ruling out that NASA spending decisions simply followed local private sector trends. Our results are also robust to controlling for industry-specific trends, military contracting, and skill.

Space race spending was economically large so we might expect local effects through a fiscal multiplier channel even without technological spillovers. We compare the fiscal multiplier for NASA contractor spending implied by our estimates to the literature to get a sense of this. Our results imply a localized NASA contractor fiscal multiplier of about 0.3 during the space race period, as measured by changes

in manufacturing value added, and a fiscal multiplier of 0.4 in the post–space race period. Ramey's (2019, p. 103) survey of the literature concludes that "most estimates of government spending multipliers for general categories of government spending for averages over samples are in the range of 0.6 to 0.8, or perhaps up to 1." Perhaps related to space spending during the Cold War, economists have estimated multipliers in the range of 0.3–1.2 from exogenous military spending triggered by wars or foreign policy initiatives.[1] Thus, we find that R&D contractor spending during the Cold War space race had a relatively small impact on economic output when compared to typical government expenditures or even defense-related spending.

Third, we estimate localized productivity spillovers from NASA contractor spending. If technologies discovered by NASA contractors spilled over to neighboring firms, then we would expect local productivity enhancements. In addition, new technologies may take time to diffuse, so NASA contractor spending may have increased local productivity into the short- or long-term. Our analysis does not detect such local technological spillovers, however. One important caveat is that our estimates are likely to be lower bounds for technological spillovers from NASA as they do not account for other types of NASA spending (e.g., at universities or at NASA's own research centers), international technology diffusion, or any effects that may have accrued outside of the manufacturing sector.

Our estimated multiplier effects based on manufacturing value added reflect local rather than national effects. Local estimates would overstate national effects if workers migrated from other locations toward places that experienced windfall NASA activity. Thus, our fourth set of results explores migration responses and implications. We turn to patent data where we build on recent advances in identifying specific inventors (Akcigit et al. 2022) to construct a patent inventor-level panel dataset. Our analysis examines whether inventors migrated toward industry-county pairs that had the ex ante capabilities to accomplish the R&D work of the space race. The results reveal that inventors working in space industries did in fact migrate toward these space locations, and the results are robust to typical county-to-county migration patterns and state tax policies.

While these migration responses would imply that the national effects of the space program would be smaller than the localized effects, other positive spatial spillovers—demand and technology being two notable examples—can counteract them. We develop a spatial framework based on Donaldson and Hornbeck (2016) that allows for workers and firms to respond to local shocks through adjustments in migration, trade, and production. Our framework accounts for multiple sources of spatial spillovers from NASA R&D to obtain the net effect of nonlocal NASA activity. Applying this theoretical framework, our fifth set of results shows that in the medium-term and long-term, overall market effects were small enough not to amplify or attenuate the positive local effects from space race activity. The implication of these findings is that the local and national fiscal multipliers associated with NASA contractor spending were largely the same.

We believe that our analysis of the space race makes important new contributions to the economics of innovation literature. A recent literature has sought to obtain

[1] See Batini et al. (2014, pp. 4–5) for a summary of this defense-related research.

causal estimates of the effect of public R&D on knowledge production (Azoulay et al. 2019; Myers and Lanahan 2022; Gross and Sampat 2023) and productivity (Moretti, Steinwender, and Van Reenen 2025).[2] Perhaps most closely related to our work here is Schweiger, Stepanov, and Zacchia (2022) who show that science cities created in Soviet Russia for space and military purposes are more productive and innovative today. We contribute to this literature by providing causal estimates of the effect of public R&D on long-term economic growth.

Second, our analysis contributes to the literature on industrial policy. Recent work has emphasized that temporary management practice transfers (Giorcelli 2019; Bianchi and Giorcelli 2022), trade protection (Juhász 2018), or university funding (Kantor and Whalley 2014, 2019; Hausman 2022; Andrews 2023) can have long-term effects on directly targeted firms or regions. Direct causal evidence on the impacts of industrial policy in Criscuolo et al. (2019) shows contemporaneous effects on employment for small firms, but the long-term effects were not considered. We complement work showing that large-scale industrial policy in South Korea during the 1970s had persistent effects on economic development and welfare (Lane 2025; Choi and Levchenko 2025). Our analysis provides new empirical insights into the spatial and temporal lags associated with public R&D that directly engaged private firms.

Third, we connect to the literature on government spending multipliers.[3] Our findings complement Ramey's (2021) work on short- versus long-term effects of public infrastructure and the work of Antolin-Diaz and Surico (2025) on the short- versus long-term effects of public spending. We also contribute to the debate on whether local fiscal multipliers adequately reflect nationwide multipliers (Nakamura and Steinsson 2014; Chodorow-Reich 2019; Ramey 2019). Our estimates of individual migration responses to local space race activity build on recent work using patent inventor panel data to understand migration responses to tax policy and their implications (Moretti and Wilson 2017; Akcigit, et al. 2022). We show that while individual patent inventors did migrate toward areas experiencing persistent fiscal shocks during the Cold War, migration effects were not sufficiently large to generate a wedge between local and national fiscal multipliers.

Modern commentators contend that space race research had particularly high returns because NASA's organization was highly effective at research coordination, and the intrinsic geopolitical motivation encouraged scientists to exert high levels of effort (Mazzucato 2021). Those advocating for significant government spending to jump-start innovation and economic growth often call for a new "Sputnik moment," harkening back to a time when the United States devoted significant treasure to racing the Soviet Union to the moon (Gruber and Johnson 2019).[4] Yet surveys of space scientists shortly after the space race suggest that NASA's role in technological development was mostly incremental (Robbins, Kelley, and Elliott 1972), and some economists since Fogel (1966)—who was writing in real time during the space race—have

[2] There is a long-standing literature that has sought to estimate social effects of R&D from case studies, regression analyses, and macroeconomic models. See Jones and Summers (2022) for a literature review.

[3] See Chodorow-Reich (2019) and Ramey (2019) for recent surveys.

[4] For example, President Joe Biden initiated his Cancer Moonshot in February 2022, renewing the effort that President Barack Obama began in 2016. But the proverbial moonshot ambition with regard to cancer is long-standing. In advocating for the National Cancer Act, President Richard Nixon argued in his 1971 State of the Union address that "The time has come in America when the same kind of concentrated effort that split the atom and took man to the moon should be turned toward conquering this dread disease."

expressed skepticism that commercially relevant technology would be developed from mission-oriented R&D.[5] While the intellectual roots of the economics of innovation draw on the proverbial "moonshot" (Nelson 1959), a measure of the effects of such large-scale public expenditures still remains elusive (Bloom, Van Reenen, and Williams 2019).[6] While our estimates imply that iconic moonshot R&D had first-order effects on economic growth in space sectors, the magnitude of the effect is on the low end of typical government spending fiscal multipliers and thus indicates that the moonshot's role in broad-based productivity growth was rather limited.

I. Historical Background

A. *The Origins of NASA*

The space race effectively began with the Soviet launch of Sputnik on October 4, 1957. The US government had intelligence that a launch was imminent (Logsdon 1995, p. 329), but the high-profile failure of the US's initial satellite effort—Project Vanguard—on live TV on December 6, 1957, instilled public fear (Divine 1993). Perceived American technological inferiority brought immediate national security concerns, as President Eisenhower emphasized in his 1958 State of the Union address: "[W]hat makes the Soviet threat unique in history is its all-inclusiveness. Every human activity is pressed into service as a weapon of expansion. Trade, economic development, military power, arts, science, education, the whole world of ideas—all are harnessed to this same chariot of expansion. The Soviets are, in short, waging total cold war."

In response to the emerging geopolitical tension, the Eisenhower administration proposed NASA in 1958, which would bring space activities under civilian control except as they related to weapons systems, military operations, and national defense.[7] The immediate need was to respond forcefully to Sputnik and to the national realization that the United States was slipping behind the Soviet Union technologically.

B. *Growth and Organization*

While Eisenhower's early efforts may have "ensure[d] that the United States remain *a* leader, not *the* leader in space, [he] did not commit the nation to an all-out race" (McDougall 1985, p. 172; italics in original). President Kennedy, however, laid down a bold marker, announcing the following on May 25, 1961, shortly after Alan

[5] Over 60 years ago, Nelson (1959, p. 297) laid bare in rather subdued language the challenge to economists to begin understanding the impacts and tradeoffs associated with national spending on scientific research: "Recently, orbiting evidence of un-American technological competition has focused attention on the role played by scientific research in our political economy. Since Sputnik it has become almost trite to argue that we are not spending as much on basic scientific research as we should … it seems useful to examine the simple economics of basic research. How much are we spending on basic research? How much should we be spending? Under what conditions will these figures tend to be different?"

[6] Business R&D appears to be shifting away from basic research (Arora, Belenzon, and Sheer 2021). In such an environment, the importance of public funding for basic research may be increasing.

[7] Military applications of space technology were to be developed by the Advanced Research Projects Agency, which was also established in 1958.

Shepard's successful suborbital space flight: "I believe that this nation should commit itself to achieving the goal, before this decade is out, of landing a man on the moon and returning him safely to the Earth." [8] Of course, the United States was nowhere close to having the technological capability to immediately fulfill that mission, so Kennedy's proverbial moonshot required a massive investment in space technology and hardware. NASA's budget grew accordingly from roughly US$(2021)7 billion (or about 0.9 percent of all federal spending at the time) in 1961 to a peak of about US$(2021)51 billion (or 4.4 percent of the federal budget at the time) in 1966.[9]

The National Aeronautics and Space Act of 1958 gave NASA broad powers to develop, test, and operate space vehicles and to make contracts for its work with individuals, corporations, government agencies, and others (Rosholt 1966, p. 61). NASA, from its inception, made the decision to contract out much of the R&D work to private contractors. T. Keith Glennan, the first NASA administrator, was an advocate for contracting out not only because of his philosophical aversion to expanding the government payroll but also because "by spreading its wealth to contractors, NASA would not just be putting together a national team to beat the Soviets in the space race but would also be invigorating the aerospace industry and strengthening the country's economy" (Hansen 1995, pp. 82–83).[10] This emphasis is reflected in the growth in personnel. While in-house NASA employees grew from 10,200 in 1960 to 34,300 in 1965, employment by NASA contractors increased from 30,500 in 1960 to a peak of 376,700 in 1965. This massive increase in space-related employment outside of NASA was concentrated in private sector contractors, which accounted for 90 percent of total NASA employment in 1965. Universities, on the other hand, accounted for only 1.7 percent of total NASA employment in 1965 (Van Nimmen and Bruno 1976, p. 106). By 1988 total NASA employment was only a fraction of its heyday, with a total workforce of 52,224 and with 56 percent of them employed by contractors (Rumerman 2000, p. 468).

C. *NASA Contractors*

While the space program required scientists and engineers to solve basic scientific questions, in practical terms winning the space race and achieving successes in subsequent space missions meant developing and engineering actual products. According to an input-output table constructed for NASA expenditures for the 1967 fiscal year, the top five manufacturing sectors accounted for about half of NASA expenditures (Schnee 1977, p. 65).[11] Similarly, relatively few firms were so-called prime NASA

[8] https://www.jfklibrary.org/learn/about-jfk/historic-speeches/address-to-joint-session-of-congress-may-25-1961.

[9] In nominal terms, NASA's budget was $744 million in 1961 and $5.933 billion in 1966. NASA's spending did decline after the landing on the moon was successfully accomplished in 1969, but it still accounted for 1.92 percent of federal spending in 1970. Subsequently, the level of spending fluctuated between 0.75 percent and 1 percent of the federal budget from 1975 until the end of the twentieth century. To provide some perspective on the magnitude of NASA's budget during the space race, consider that in 2020 the total of all nondefense federal R&D amounted to 1.5 percent of the federal budget.

[10] For further elaboration on Glennan's views see Hunley (1993, p. 5) and Dunar and Waring (1999, p. 64).

[11] The five Standard Industrial Classification (SIC) three-digit industries with the largest share of NASA spending were aircraft and parts (SIC = 372), electrical equipment (SIC = 361–366), computer and office equipment (SIC = 357), industrial inorganic chemicals (SIC = 281), and instruments (including professional and scientific) for measuring, testing, analyzing, and controlling (SIC = 381–387).

contractors. In 1965, for example, the top ten contractors received nearly 70 percent of the contract spending. Leading technology companies receiving NASA projects included North American Aviation, Boeing, Grumman Aircraft Engineering, Douglas Aircraft, General Electric, McDonnell Aircraft, International Business Machines, and the Radio Corporation of America (Van Nimmen and Bruno 1976, p. 197).

Rosholt (1966, p. 272) notes in his administrative history of early NASA work that "[t]he geographic distribution of NASA contracts was a touchy political problem. Congressmen were sensitive to the fact that most of NASA's procurement dollar was spent in a handful of states. NASA's answer was that the competence of a contractor rather than his location was the basis for awarding contracts." After all, excellence was demanded because, quite literally, lives were at stake. Dieter Grau, the Director of the Quality and Reliability Assurance Lab at the Marshall Space Flight Center, put the logic simply: "[Y]ou cannot put a man on a [launch vehicle] and say 'if it fails, and if you get killed, take the next one.'" Marshall, therefore, demanded that contractors shift from their perhaps existing "mass production with acceptable errors" mentality to one where "craftsmanship-do it right the first time-with no error" was the imperative (Dunar and Waring 1999, p. 45).

D. *Technology Impacts*

Winning the space race did not necessarily entail developing entirely new technologies as much as combining or speeding along the development of existing technologies (Robbins, Kelley, and Elliott 1972). NASA's mission-oriented objective, especially during the race to the moon, led to R&D breakthroughs that might cause the casual observer to wonder whether any broader economic impacts would even be expected. As examples, consider NASA patents from the space race era, including patents on a space capsule design, a navigation and guidance system, and a moon-landing apparatus. Yet the space race did produce and escalate innovative breakthroughs in several areas, such as cryogenics, integrated circuits, digital communications, and computer simulation, that had the potential to spill over more broadly (see, e.g., Bilstein 1996). Examples of burgeoning technologies in which NASA participated in enough fashion that the agency considered them spinoffs include magnetic resonance imaging, remote sensing, a gas analyzer, and a circuit connector.[12]

II. Data Construction and Descriptive Statistics

This paper uses newly constructed datasets on technological specialization, space sector activity, and manufacturing during the Cold War era. Our measurement relies on three components: (i) declassified CIA intelligence documents detailing Soviet space capabilities, which are then matched to pre-Sputnik US patents, thus enabling us to identify space sector industry-county pairs based on technological similarity; (ii) industry-county-level NASA contractor spending data that are used to measure space sector activity and patents to measure innovation outcomes; and

[12] See https://spinoff.nasa.gov/ for NASA's profile of over 2,000 spinoff technologies for which the agency claims credit.

(iii) industry-county-level manufacturing census data used to measure outcomes in the real economy. In this section, we briefly describe the construction of these components and some data limitations. Detailed discussions of the construction of each variable, as well as the data sources, are available in the Supplemental Appendix, Sections 1 and 2.

A. *Space Technologies and Space-Capable Places*

Our research design compares changes in outcomes between industry-county combinations that specialized in research forming the building blocks of spaceflight technology before the space race to those that did not. We first need to measure which technologies were the building blocks of spaceflight technology. At first glance, using observed NASA technology choices might seem a promising approach. However, NASA's technological choices reflect both mission requirements and opportunities provided by US leadership in specific technologies that could help win the race to the moon. Locations that specialized in technologies where the United States had technological superiority—and selected by NASA for that reason—may have been poised for growth regardless of the space program. Because NASA may have simply harvested technological potential rather than having developed technological breakthroughs to solve emergent challenges, a correlation between NASA activity and growth may not reflect a causal effect.

To address this issue, we define the building blocks of spaceflight technology from Soviet technology choices. Soviet choices did not necessarily reflect the scientific areas where the United States had technological superiority, as a lack of US-Soviet trade or knowledge sharing made them irrelevant. Instead, Soviet technological choices reflected mission requirements as well as opportunities provided by Soviet leadership in specific technologies. We obtain these technologies by digitizing the CIA's declassified national intelligence estimates of Soviet space capabilities from 1947 to 1991.[13]

We classify the industry-county pairs with regard to pre–space race spaceflight technology by searching for post-Sputnik Soviet spaceflight technologies in the US patent record prior to the launch of Sputnik in 1957. Using text similarity to connect units in technology space has been shown to quantify economically meaningful concepts (see, e.g., Azoulay et al. 2019; Myers 2020; Myers and Lanahan 2022). To estimate a numerical similarity score between each CIA document and each US patent we use term frequency cosine similarity for a set of scientific terms. Our textual similarity measure captures spaceflight technological similarity regardless of how patents were classified by the US Patent and Trademark Office (USPTO). Examples of patents that are highly similar to a specific CIA document are shown in Figure 3. We see patents dealing with pop-up fins, orbital devices, and satellites.[14]

[13] The titles and dates of the declassified CIA intelligence documents are provided in Supplemental Appendix Table A1. Our primary space capability measure is based on the post-1958 documents as these are more likely to have an exclusive focus on space.

[14] Examples of Science Direct (SD) technology terms most frequent in patents owned or funded by NASA, shown in Supplemental Appendix Table A2, include "aircraft," "antennae," and "propellant." Examples of SD technology terms most frequent in the CIA's Soviet space technology intelligence reports, shown in Supplemental Appendix Table A3, include "missiles," "satellites," and "orbitals." Supplemental Appendix Table A4 reports the

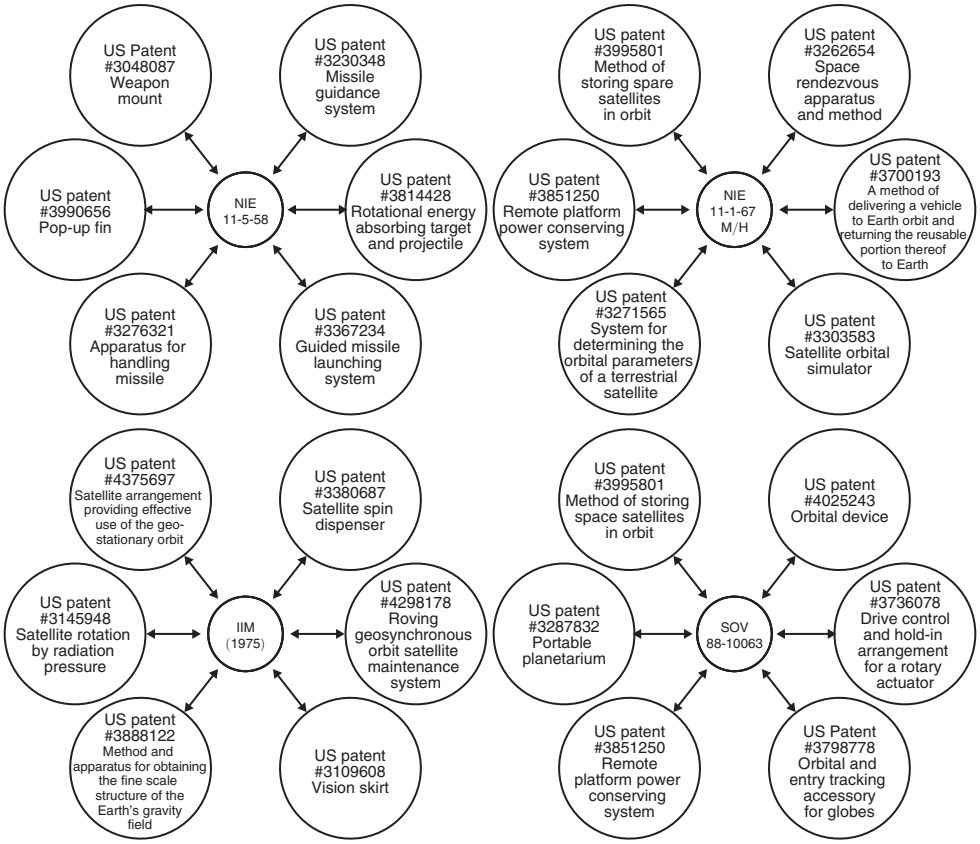


FIGURE 3. PATENTS HIGHLY SIMILAR TO NATIONAL INTELLIGENCE ESTIMATES OF SOVIET SPACE CAPABILITIES

Notes: Authors' calculations using declassified Central Intelligence Agency (1958–1992) documents and patent data from 1945 to 1958. Each figure lists the patents with technologies most similar to the indicated CIA document. See Supplemental Appendix Table A1 for a list of the relevant intelligence documents.

We aggregate these textual similarity measures across all pre-1958 US patents in an industry-county cell to create our "space capability" measure. We discuss our approach in detail in Supplemental Appendix Section 2.2.

B. *NASA Contractor Spending and Patents*

We measure NASA activity using expenditures and patents. We collect and digitize new data on NASA primary contractors from NASA's historical databooks. These data include the company names, amount of primary contracts, and place of performance (in addition to location of company headquarters) for the top 100

SD terms occurring frequently in both CIA and patent documents. Terms such as "aircraft," "spacecraft," and "satellites" are frequently found in both types of documents.

contractors from 1963 to 1992.[15] NASA primary contracts, in practice, flowed to a small number of large firms so that the top 100 firms accounted for between 87 percent and 92 percent of total contractor spending. Moreover, NASA contractor spending was highly concentrated in two space sectors—transportation equipment and electronics equipment—accounting for nearly 90 percent of NASA manufacturing contractor spending.[16]

A second source we use to measure NASA activity is patents owned or funded by the agency. For patents prior to 1976, this information is drawn from Fleming et al. (2019) who have scraped assignee and government funding information from the full text of USPTO patents. After 1976 the information is directly reported by the USPTO. We allocate granted patents to locations. We utilize a few sources to obtain a county for each patent. For the data before 1975 we use the HistPat database that has scraped the full text of the patent to assign each patent to the most appropriate county (Petralia, Balland, and Rigby 2016). For the post-1975 data we use the USPTO Patentsview data that have the exact address for each inventor. For patents with multiple inventor locations we assign a proportional fraction to each location.[17]

An important limitation with using patent data to measure government-sponsored innovation is that before the 1980 Bayh-Dole Act began the process of creating a uniform patent policy, different agencies had different assignment and reporting policies, in a way that matters for measurement.[18] When NASA was created in 1958, the founding legislation gave the government all rights to the inventions made within NASA programs, but the administrator had the discretionary ability to waive such rights and grant contractors ownership of their intellectual work product.[19] Thus, NASA patents may measure NASA activity with significant error if in fact contractors were able to successfully lobby for waivers.[20] For this reason, we measure NASA activity using both NASA contractor spending and NASA patents.[21]

[15] Companies receiving the largest amount of NASA contracts include Boeing, Ford, General Motors, General Electric, Grumman, IBM, McDonnell Douglas, and North American Aviation. Prominent metro areas containing counties having high levels of NASA spending include Los Angeles (Los Angeles County, CA), New York City (Nassau County, NY), and Cincinnati (Hamilton County, OH).

[16] Figure 2, panel A shows the distribution of average NASA contractor spending across industries.

[17] We build a crosswalk between Federal Information Processing Standard (FIPS) counties and city-state name text fields from the USPTO patent technology team database (https://bulkdata.uspto.gov/data/patent/ptmtdvd/). This database assigns each address on a patent from 1969 to 2014 to a fips county. Most city-state text fields are assigned to a unique location. For the few that are not, we assign the city-state text to the largest county listed.

[18] We thank an anonymous referee for alerting us to this issue.

[19] See McDougall (1985) and Kraemer (2001). Eisenberg (1996) discusses the legal tradeoffs between so-called "title policy" versus "license policy" that were inconsistently utilized across federal government agencies.

[20] Jaffe, Fogarty, and Banks (1998, pp. 188–9) note that the NASA waiver policy became increasingly lenient through the 1970s, and by the early 1980s waivers were "essentially automatically granted." The Bayh-Dole Act of 1980 eliminated the need for universities, nonprofit institutions, or small businesses to apply for formal waivers from NASA. President Ronald Reagan issued an executive order in 1983 that directed government agencies to extend the Bayh-Dole titling privileges over federally funded research to all government contractors, including large businesses. See Eisenberg (1996).

[21] Map A1 in the Supplemental Appendix shows which sample counties had a NASA patent or any NASA spending from 1947 to 1992. In Supplemental Appendix Table A5 we show that our patent-level space score based on textual similarity between the patent and CIA intelligence documents strongly predicts NASA ownership or funding of a patent, conditional on military funding, technological area, and county fixed effects (see Supplemental Appendix Section 2.2 for further discussion). Supplemental Appendix Table A6 shows that NASA spending and NASA patenting variables are spatially correlated, though perhaps less correlated than would be expected because of the patent-based measurement challenges noted above.

The challenge of measuring military-sponsored innovation—an important control in our analysis—using patents is likely to be significant. The army and navy were historically "license agencies" whereby patents were assigned to contractors, so we will likely undercount military-supported innovation. To address this concern, we control for a variety of measures of potential military research activity locally and develop a new measure of military patents at the industry-county-year level by searching patent texts for technologies contained in the military technology glossary, as described in the Supplemental Data Appendix.

Figure 1 plots the times-series of NASA activity from 1947 to 1992. In panel A we see that real NASA spending increased substantially after 1958. Spending peaked in 1965 at the height of the space race before declining more than 50 percent by the mid-1970s. While spending steadily increased thereafter, it never returned to the space race peak. In panel B we see that NASA patents were very low before NASA was founded in 1958.[22] During the space race the number of patents granted per year increased from 21 in 1961 to 256 in 1969. From 1967 until today the number of patents per year has fluctuated in the 150–300 range. In the postwar period the total number of patents and total number of government patents increased much more slowly and gradually than NASA's. Both NASA spending and patenting show a sharp increase in activity after the launch of the space race. NASA contractor spending fell after the peak of the space race in the mid-1960s, while patenting remained elevated.

C. *Manufacturing Data*

The primary data we use to estimate the impact of NASA research and development on value added, employment, and labor income is from the census of manufactures. We digitize data at the industry-county level from the censuses of 1947, 1954, 1958, 1963, 1967, and 1972 and combine them with existing digital sources from 1977, 1982, 1987, and 1992.[23] We obtain data on total value added, total employment, total annual wages, and total plant and equipment additions for each industry-county cell. We use two-digit SIC industries (1972 definition) in the county as the unit of analysis.[24]

D. *Additional Data*

We also employ data on local measures of skill from the population census, the number of research scientists from the National Register of Scientific and Technical Personnel, the number of IBM mainframes installed in various locations, defense spending, and transportation cost data. Details of the construction and source of each variable are described in the Supplemental Appendix.

[22] The few patents from before 1958 are likely from patents under NASA's precursor, the National Advisory Committee for Aeronautics. The patents were later reassigned to NASA (Ferguson 2013).

[23] Manufacturing census data are available at the industry-county level after 1992; however, the data are reported at the North American Industry Classification System (NAICS) instead of SIC level from 1997 onward. For this reason and given our focus on the space race prior to the end of the Cold War, we do not examine later years of data.

[24] The census manufacturing data are also available at the three- and four-digit SIC $\times$ county level. We choose the two-digit level, however, because the masking of cells with few establishments results in extensive missing data if we were to use disaggregated data. Using two-digit level data results in fewer nonreported observations.

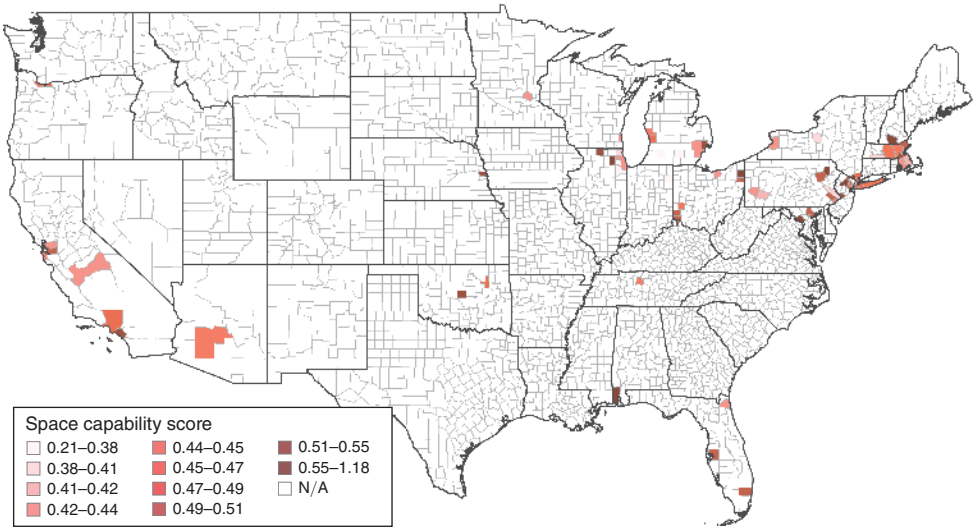


FIGURE 4. SPACE CAPABILITY SCORES OF SPACE-ACTIVE COUNTIES IN 1958

Notes: The map displays county-level averages of space capability scores for the urban counties within defined MSAs that had manufacturing activity in the space sector in 1958 and that consistently reported manufacturing throughout our sample period (i.e., 1947–1992). The space capability score is detailed in Section 2.2 of the Supplemental Appendix.

E. *Sample Selection and Descriptive Statistics*

The sample of counties and industries represented in our analysis is based on those reported in the census of manufactures, with the caveat that we exclude the few counties that had no patents between 1945 and 1958 or those that are not in a metropolitan statistical area (MSA).[25] Effectively, our sample captures the major urban labor markets that had innovative activity prior to 1958. Entry and exit of specific manufacturing sectors in a county leads to an unbalanced panel. Data may also be unreported because the number of establishments was below the threshold for confidentiality. We require that an industry-county cell report in the 1958 census and in at least eight censuses to address issues that might arise with a highly unbalanced sample. Additional sample restrictions include a requirement that both value added and employment were reported and that one of the industry-county cells within a county is space related and that the county belongs to an MSA. We also drop the observations that appear in North Dakota, South Dakota, or Wyoming because only a single county in each state reported manufacturing data. Our analysis sample contains 6,759 industry-county observations from 86 counties and 19 two-digit SIC industries from 1947 to 1992.

Figure 4 shows the spatial distribution of space capability scores for the sample counties. The map displays county level averages for the urban counties within

[25] We exclude these counties without pre-1958 patents because we are unable to compute a space capability score for them. We exclude those without an MSA as we cluster our standard errors at the MSA level.

TABLE 1—DESCRIPTIVE STATISTICS OF PRE–SPACE RACE ERA

	All	Space capability score$_{ij<1958}$		Difference	Space industry$_j$		Difference	Difference
		≥ Median	< Median	(2)–(3)	Yes	No	(5)–(6)	(4)–(7)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Manufacturing census data								
Value added$_{ij}$ ($1958 million)	72	81	64	0.205	113	67	0.116	0.352
	(153)	(177)	(123)		(275)	(131)		
Employment$_{ij}$ (1958)	7,760	8,571	6,947	0.232	12,044	7,266	0.110	0.177
	(15,789)	(17,374)	(13,997)		(27,739)	(13,712)		
Labor income$_{ij}$ ($1958)	4,832	4,914	4,749	0.017	4,981	4,814	0.016	0.496
	(840)	(771)	(898)		(746)	(850)		
Capital investment$_{ij}$	3,572	4,125	3,016	0.163	4,126	3,508	0.443	0.389
($1958 '000's)	(8,941)	(11,435)	(5,345)		(9,436)	(8,887)		
Panel B. Patent data								
Total patents$_{ij}$ (1953–1958)	60	71	49	0.050	127	52	0.000	0.812
	(137)	(161)	(109)		(217)	(123)		
Navy patents$_{ij}$ (1953–1958)	0.03	0.05	0.00	0.007	0.14	0.01	0.032	0.034
	(0.21)	(0.29)	(0.06)		(0.53)	(0.13)		
Army patents$_{ij}$ (1953–1958)	0.01	0.02	0.01	0.325	0.03	0.01	0.461	0.340
	(0.12)	(0.13)	(0.10)		(0.16)	(0.11)		
Panel C. Population census and other county data								
Population$_i$ (1960)	1,003,562	1,063,508	943,452	0.320				
	(1,193,431)	(1,287,886)	(1,089,000)					
High school graduate percent$_i$	44	45	43	0.002				
(1960)	(8)	(8)	(8)					
Research scientists$_i$ (1962)	5,063	5,737	4,388	0.015				
	(6,048)	(6,416)	(5,583)					
IBM mainframe computers$_i$	2.94	3.29	2.58	0.054				
(1961)	(3.95)	(4.17)	(3.70)					
Industry-county observations	735	368	367		76	659		

Notes: See the Supplemental Data Appendix, Sections 1 and 2, for variable definitions, construction, and sources. The space capability score is discussed in Section 2.2 of the Supplemental Appendix. The unit of observation is county × two-digit SIC industry in panels A and B, and county in panel C, where i and j index county and industry, respectively. In columns 1, 2, 3, 5, and 6 the main entries are means for the variables indicated with standard deviations in parentheses. Column 4 reports the p-value for the hypothesis test that the values in 2 and 3 are different. Column 7 reports the p-value for the hypothesis test that the values in 5 and 6 are different. Column 8 reports the p-value for the hypothesis test that the values in 4 and 7 are different. The table represents data for the full sample from 1958.

defined MSAs that had manufacturing activity in the space sector in 1958 and that consistently reported manufacturing throughout our sample period (i.e., 1947–1992). The map shows that many space places—that is, those with a relatively high space capability score—were distributed throughout the country, with a small amount of clustering in the northeast. In Section IV we show econometrically that our measure of pre-Sputnik space-related research performs well in explaining how and where NASA subsequently allocated its spending.

Table 1 provides a first look at summary statistics of relevant measures in 1958, the first year immediately after Sputnik was launched. Column 1 presents the means and standard deviations of key variables for the full sample. We first stratify industry-county pairs based on their level of pre-1958 space relevant technology capability. In columns 2 and 3 we stratify based on whether an industry-county had an above or below median space capability score as defined above. Column 4 reports the p-value for differences in the baseline variables for the full sample. In columns 5–7 we conduct the same analysis where we stratify by whether the

industry was a space industry or not. In column 8 we report *p*-values for the difference between the differences in the baseline values.

Columns 1–4 show that industry-counties that would later be more exposed to the space race were quite similar to nontreated counties in 1958. In columns 2 and 3 we see that those locations that were eventually more heavily exposed to the space race generally had higher average labor income in manufacturing, more navy patents, and higher skill but were otherwise quite similar in other manufacturing outcomes, patents, population, and skill measures. The results in columns 5–7 show similarly that only baseline differences in manufacturing average labor income and total patents existed across space and non-space industries. Turning to column 8 we only see statistically significant baseline differences between the two differences for navy patents. Table 1 provides evidence that triple difference treatment and control industry-county pairs were quite balanced in 1958 before the United States embarked on the space race.

III. Local Effects of Public R&D

A. *Conceptual Framework*

Space spending in a location could affect manufacturing output either through a local fiscal multiplier or through technological spillovers that enhanced productivity within the target industry or colocated industries. To the extent that NASA spending contributed to local economic growth, one of the goals of our empirical analysis is to parse the productivity contributions from the more standard fiscal multiplier effects. Furthermore, the manifestation of the economic effects of space activity in an industry-county could accrue over time or across locations, which our empirical analysis also seeks to quantify. With respect to spatial lags in the effects of the space economy in a specific location, the impact on other regions could be two-fold. On the one hand, the effect on neighbors could be positive if the space-stimulated regions demanded goods and services of their neighbors or if they acquired manufacturing productivity gains associated with their neighbors' space activity. On the other hand, if labor migrated from neighboring areas to space-active areas, then that could have had a deleterious effect on neighbors' economies. The magnitude of these potential gains or losses accruing to neighbors will help to determine how well the local multiplier we calculate represents the overall impact of NASA space spending on the broader economy.

B. *Empirical Approach*

This section presents our main approach and results. We analyze how the launch of the space race in 1958 affected a variety of activities in relatively high space-capable industry-county pairs—that is, industries within places that prior to Sputnik had specialized in technologies that would later prove useful for winning the space race. For this analysis we use data on NASA expenditures and patenting and manufacturing outcomes in the census years of 1947, 1954, 1958, 1963, 1967, 1972, 1977, 1982, 1988, and 1992.

We test whether NASA resource allocation and manufacturing disproportionately grew in industry-county cells that specialized in the early building blocks of space research before the space race even began. We estimate our triple difference model using the following equation:

$$
\begin{aligned}
(1)\quad Y_{ijt} = {} & \beta_1 + \beta_2 \textit{High Space Capability}_{ij<1958} \times \textit{Space Race}_t \\
& + \beta_3 \textit{High Space Capability}_{ij<1958} \times \textit{Post-Space Race}_t \\
& + \beta_4 \textit{High Space Capability}_{ij<1958} \times \textit{Space Race}_t \times \textit{Space Industry}_j \\
& + \beta_5 \textit{High Space Capability}_{ij<1958} \times \textit{Post-Space Race}_t \\
& \times \textit{Space Industry}_j + \textit{Total Pre-1958 Patents}_{ij} \times \gamma_t + \delta_i + \theta_j \\
& + \gamma_t + \upsilon_{ijt}.
\end{aligned}
$$

The outcome variables (Y_{ijt}) are NASA activity and manufacturing activity measures, in county i, industry j, and year t. $\textit{High Space Capability}_{ij<1958}$ is a binary variable that takes a value of one when the text similarity between technologies mentioned in pre-1958 patents in county i, industry j, and those mentioned in the post-1958 CIA document is above median. $\textit{Space Race}_t$ is dummy variable that takes a value of one during the space race (i.e., 1959–1972, inclusive) and zero otherwise. $\textit{Post-Space Race}_t$ is a dummy variable that takes a value of one after the space race (i.e., 1973–1992, inclusive) and zero otherwise. $\textit{Space Industry}_j$ is a dummy variable that takes a value of one if industry j is a space industry (i.e., transportation or electronics) and zero otherwise. δ_i is a full set of county fixed effects, θ_j is the full set of industry fixed effects, and γ_t is a full set of year effects.

As industry-counties with pre-1958 space specialization might have had unobserved time-invariant characteristics that drove space activity before, during, and after the space race, we include both industry and county fixed effects in our analysis. We include *Total Pre*-1958 $\textit{Patents}_{ij} \times \gamma_t$ controls to account for differential trends based on the pre-existing level of patenting in an industry-county. In other versions of the model, we include MSA $\times$ year fixed effects to flexibly control for MSA-level trends. To account for potential correlation of shocks within MSAs across time and within industries across time, we two-way cluster standard errors at the MSA $\times$ industry level.

Our coefficients of interest are β_4 and β_5. We expect them to be positive in the NASA expenditure estimation since industry-county pairs that were specialized in space-relevant technologies before 1958 were likely to experience more NASA activity after 1958 once the space race began. We expect the coefficients to reflect larger effects during the space race than after as NASA scaled down after the successful moon landing. For the manufacturing estimation, we expect the coefficients to be positive if NASA expenditure generated growth. Whether β_4 or β_5 would be larger for manufacturing depends on what benefits the space race activity provided. If NASA spending primarily acted as a government expenditure shock, any fiscal multiplier effects would decline as NASA scaled down after the space race.

Alternatively, for a technological shock where spillovers took time to manifest, then the measured effects would be expected to grow over time.

Our research design is based on the idea that industry-counties that specialized in scientific research before 1958, which ultimately became important space technology areas after 1958, did not experience higher levels of NASA activity until after the space race began. We regard this assumption as plausible given that the decision to go to the moon was only made after the launch of Sputnik in 1957. In addition, as space funding was highly targeted toward just a few sectors—Figure 2, panel A shows that two sectors (transportation and electronics) accounted for 89 percent of contractor funding—our research design harnesses the industry-specific nature of the shock.[26] As NASA did not even exist until 1958, we cannot examine pre–space race trends for NASA expenditure. We do, however, examine the possibility that NASA may have allocated space funding in response to pre-existing trends in the industry-county manufacturing sector in later analyses.

C. *NASA Contractor Spending and Patents*

The results of estimating equation (1) using NASA contractor spending and NASA patents as outcome variables are reported in Table 2. Panel A focuses on the extensive margin, so any NASA spending greater than zero is coded as one, but otherwise the dependent variable is zero. Panel B shows estimates using a Poisson pseudo-maximum likelihood (PPML) model. Across the extensive and intensive margins, we see in columns 1 and 2 that NASA contractor spending both during (1958–1972) and after (1973–1992) the space race was larger in industry-county pairs that had previously attained the expertise to conduct NASA work. Our preferred estimates in panel B, column 1 imply that NASA spent US$(1958) 264 million more during and US$(1958) 240 million more after the space race in industry-counties with a relatively strong prior history in space-related research. We use these magnitudes to estimate local fiscal multiplier effects below.

In columns 3 and 4 we report results that use patents owned or funded by NASA as the outcome variable. The positive and precise point estimates in panel A are consistent with the NASA spending results. They differ in that the post–space race effects are more than double those during the space race.[27] The PPML results do not reveal a positive statistically significant impact on the intensive margin, however.

Differences in effect dynamics between NASA contractor spending and NASA patenting, as revealed in panel A, could indicate that it takes time for contractor activity to translate to new innovations. This finding is consistent with a contemporary assessment of the technological developments that occurred during the space race (see Robbins, Kelley, and Elliott 1972). An important caveat for this interpretation is

[26] More specifically, our triple-difference approach captures the essence of NASA's decision-making when allocating funds. NASA's spending choices with regard to industry sectors were determined by the demands of the space mission, not simply by the availability of existing space-related technologies that might have been available in other sectors that would not have been helpful to the mission. Quite simply, transportation and electronics were required for a successful mission to the moon. Those suppliers with comparative advantage in space-capable technologies within this so-called space sector were ultimately selected by NASA to produce goods to help achieve the mission.

[27] Supplemental Appendix Figure A1 shows the dynamics effects for an annual series of NASA patents. There are no pre-trends evident in the series.

TABLE 2—SPACE CAPABILITY, NASA SPENDING, AND NASA PATENTS: EXTENSIVE MARGIN AND PPML APPROACHES

	NASA spending$_{ijt}$		NASA patents$_{ijt}$	
Dependent variable	(1)	(2)	(3)	(4)
Panel A. Extensive margin				
High space capability$_{ij<1958}$ × space race$_t$	0.00	0.00	0.01	0.01
	(0.01)	(0.01)	(0.01)	(0.01)
High space capability$_{ij<1958}$ × post-space race$_t$	0.00	0.00	0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)
High space capability$_{ij<1958}$ × space race$_t$ × space industry$_j$	0.07	0.07	0.05	0.05
	(0.01)	(0.02)	(0.02)	(0.02)
High space capability$_{ij<1958}$ × post-space race$_t$ × space industry$_j$	0.08	0.08	0.12	0.12
	(0.02)	(0.03)	(0.03)	(0.03)
R^2	0.15	0.19	0.26	0.38
Observations	6,759	6,759	6,759	6,759
Panel B. PPML				
High space capability$_{ij<1958}$ × space race$_t$	−33.49	−83.34	0.01	0.00
	(8.86)	(19.65)	(0.05)	(0.08)
High space capability$_{ij<1958}$ × post-space race$_t$	−82.06	−94.02	−0.05	−0.09
	(39.02)	(89.54)	(0.03)	(0.06)
High space capability$_{ij<1958}$ × space race$_t$ × space industry$_j$	264.24	434.47	0.03	0.04
	(32.19)	(65.14)	(0.06)	(0.08)
High space capability$_{ij<1958}$ × post-space race$_t$ × space industry$_j$	240.17	377.48	0.02	0.04
	(50.74)	(128.70)	(0.05)	(0.09)
Pseudo R^2	0.94	0.95	0.71	0.73
Observations	1,093	656	5,244	3,115
County fixed effects	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y
Pre-1958 patents$_{ij}$ × year fixed effects	Y	Y	Y	Y
MSA × year fixed effects		Y		Y

Notes: See the Supplemental Data Appendix, Sections 1 and 2, for variable definitions, construction, and sources. The unit of observation is two-digit SIC industry × county × year, indexed by *j*, *i*, and *t*, respectively. Each column in the table reports the results from estimating one version of equation (1) in the text. *High Space Capability*$_{ij<1958}$ is an indicator variable reflecting a industry-county's being above median in terms of the similarity between the technologies present in pre-1958 patents and the CIA's intelligence estimates of Soviet space capabilities between 1958 and 1992 (i.e., the space capability score), as described in the text and the Supplemental Data Appendix. Space race years are 1963, 1967, and 1972. Post–space race years are 1977, 1982, 1987, and 1992. *Space Industry*$_j$ is an indicator variable for industry *j* being a space industry. In panel A main entries report coefficient estimates. In panel B main entries report marginal effect estimates from a PPML model. Standard errors are two-way clustered at the MSA × industry level and are reported in parentheses. In panel A the dependent variable is an indicator variable that takes a value of one if a positive value of *NASA Spending*$_{ijt}$ is reported (columns 1 and 2) or takes a value of one if a positive value of *NASA Patents*$_{ijt}$ is reported (columns 3 and 4).

that patent attribution to NASA is measured with error, particularly during the space race era when government interest statements were not yet required for patents.[28]

[28] We thank a referee for making us aware of this issue. Using an internal list of NASA patents identified in the NASA Technical Reports Server (NTRS) after 1972, we find large agreement with the Fleming et al. (2019) measure of patents with NASA involvement and NASA's own assessment. Thus, it appears that NASA largely followed a "title policy" in that the agency seems to have retained ownership of the patents developed with its funding. Or, if the agency did license the patent to the private contractor, it appears that an explicit government interest was routinely declared.

TABLE 3—SPACE CAPABILITY AND MANUFACTURING

	log(value added$_{ijt}$)		log(employment$_{ijt}$)		log(capital$_{ijt}$)		log(TFP$_{ijt}$)	
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High space capability$_{ij<1958}$ × space race$_t$	−0.07	−0.08	−0.08	−0.09	−0.07	−0.04	0.01	0.00
	(0.07)	(0.07)	(0.06)	(0.06)	(0.09)	(0.09)	(0.03)	(0.02)
High space capability$_{ij<1958}$ × post-space race$_t$	0.03	0.03	0.05	0.06	−0.03	0.01	−0.01	−0.02
	(0.07)	(0.08)	(0.06)	(0.06)	(0.08)	(0.08)	(0.02)	(0.02)
High space capability$_{ij<1958}$ × space race$_t$ × space industry$_j$	0.35	0.38	0.42	0.45	0.31	0.25	−0.04	−0.03
	(0.06)	(0.07)	(0.08)	(0.08)	(0.13)	(0.15)	(0.03)	(0.04)
High space capability$_{ij<1958}$ × post-space race$_t$ × space industry$_j$	0.38	0.36	0.36	0.34	0.56	0.50	−0.02	−0.02
	(0.08)	(0.11)	(0.10)	(0.13)	(0.17)	(0.16)	(0.03)	(0.03)
County fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Pre-1958 patents$_{ij}$ × year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
MSA$_i$ × year fixed effects		Y		Y		Y		Y
R^2	0.66	0.68	0.53	0.55	0.40	0.46	0.85	0.86
Observations	6,759	6,759	6,759	6,759	6,759	6,759	6,759	6,759

Notes: See the Supplemental Data Appendix, Sections 1 and 2, for variable definitions, construction, and sources. The unit of observation is two-digit SIC industry × county × year, indexed by j, i, and t, respectively. Each column in the table reports the results from estimating one version of equation (1) in the text. log(*TFP*) is defined as log(A_{ijt}) from estimating the production function $Y_{ijt} = A_{ijt}K_{ijt}^{\alpha}L_{ijt}^{\beta}$ by OLS. *High Space Capability*$_{ij<1958}$ is an indicator variable reflecting an industry-county's being above median in terms of the similarity between the technologies present in its pre-1958 patents and the national intelligence estimates of Soviet space capabilities between 1958 and 1992, as described in the text and Supplemental Appendix. Space race years are 1963, 1967, and 1972. Post–space race years are 1977, 1982, 1987, and 1992. *Space Industry*$_j$ is an indicator variable for the industry j being a space industry. Main entries report coefficient estimates. Standard errors are two-way clustered at the MSA × industry level and are reported in parentheses.

D. *Manufacturing*

In Table 3 we report the main manufacturing results. The results in columns 1 and 2 show that manufacturing value added grew faster in the space sectors and counties that were predisposed to conduct the space research needed to complete the moon mission as evidenced by their pre-Sputnik patent similarity to later Soviet space research. The point estimates during the space race are similar to those for the era after the race to the moon had ended after 1972. This finding may be expected if contemporaneous NASA expenditures during the space race stimulated manufacturing activity that continued similarly with post–space race NASA contractor spending. The evidence is not strong that knowledge gained during the space race era manifested into larger long-term gains for contractors. In columns 3 and 4 we see a similar pattern of results for employment. Again, the effects are larger and more precise during the space race than after it had ended. Our results differ for capital, reported in columns 5 and 6, where post–space race effects are larger than those during the race to the moon. This outcome might be expected if capital accumulation occurs with a lag.

That the magnitudes of the value added, employment, and capital effects are quite similar may suggest little productivity effect. We measure total factor revenue-based productivity by estimating the production function $Y_{ijt} = A_{ijt}K_{ijt}^{\alpha}L_{ijt}^{\beta}$ to recover manufacturing revenue TFP at the industry-county-year level (i.e., A_{ijt}). We see no statistically significant effects of the space race on measurable productivity in columns 7 and 8. The point estimates are quite close to zero and even trend negative. Despite the caveats that this productivity measure is revenue-based and does not

account for endogenous choices of inputs, there is little evidence that a positive productivity effect or resulting technological spillovers from the space race played a role in boosting manufacturing value added.[29]

E. *Prior Trends*

A potential lingering concern is that NASA activity may have been endogenous to local outcomes. It could be the case, for example, that NASA was harvesting technologies by responding to unobserved productivity shocks within an industry-county cell. While our reading of the historical evidence indicates that NASA did not follow trends in the productivity of manufacturing firms or of specific locations because of the imperative to win the race to the moon quickly, exploring prior trends is an important specification check.

In Figure 5 we graphically present dynamic versions of our main econometric model with 1958 as the reference year.[30] The results from this analysis reveal little evidence of prior trends. The coefficients of the 1947 and 1954 interactions are very close to zero and not statistically different from zero at any conventional confidence level. These results lend additional credibility to our research design.

F. *Military Activity and Skills*

The Cold War period in the United States featured dramatic expansions in military-sponsored research and skill accumulation. Both factors may have been important for the growth of manufacturing output and potentially correlated with the rise of NASA activity itself. A simple approach to address this concern is to control for these factors at the county or preferably county × industry level.

In panel A of Table 4 we add controls for military activity. We utilize newly digitized data on government-sponsored patents in this period from Fleming et al. (2019) to measure army and navy patents at the industry-county level. Controlling for these patents in columns 1, 3, 5, and 7 of Table 4, panel A, does little to alter our estimates of NASA's effect on manufacturing.[31] Since measuring military

[29] Changes in revenue, holding constant measured inputs, have several components: changes in the quantity of output produced, changes in the quality of output produced, and changes in the quality-adjusted price. We cannot separately identify these components, so our results capture effects across all of these margins. We thank a referee for clarifying what our measure captures.

[30] The model we estimate is

$$\begin{aligned}(2)\qquad \log(Y_{ijt}) &= \alpha_1 + \sum_{\substack{k=1947\\ k\neq 1958}}^{1992} \gamma_{k1}\, High\ Space\ Capability_{ij<1958} \times Year \\ &= k_t + \sum_{\substack{k=1947\\ k\neq 1958}}^{1992} \gamma_{k2}\, High\ Space\ Capability_{ij<1958} \times Year \\ &= k_t \times Space\ Industry_j + Total\ Pre\text{-}1958\ Patents_{ij} \times \gamma_t + \delta_i + \theta_j + \gamma_t + \upsilon_{ijt},\end{aligned}$$

where $Year = k_t$ is a dummy variable that takes a value of one for manufacturing census year k and is zero otherwise. The excluded year is manufacturing census year 1958. Other variables are defined as in equation (1). Supplemental Appendix Table A7 reports the coefficients γ_{k2}, and Appendix Figure A2 displays the coefficients γ_{k1}, which are all close to zero and statistically insignificant. Not finding a main effect for space capability lends credence to our research strategy that utilizes NASA-revealed demand to define the space industry variable.

[31] This result may be expected as the spatial correlation between military patents and NASA patents turns out to be quite small. See Supplemental Appendix Table A6. Patent assignment to government agencies before the 1980

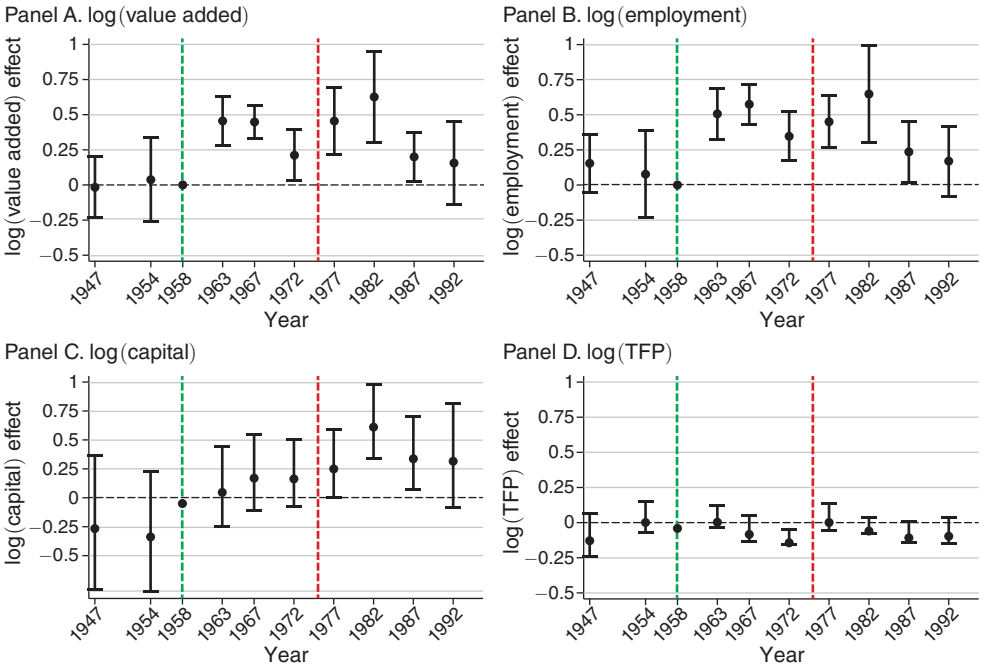


FIGURE 5. SPACE CAPABILITY AND MANUFACTURING: SPACE INDUSTRY EFFECT DYNAMICS

Notes: Each panel in the figure displays the results from estimating one version of equation (2) in the text for the outcome indicated, with 1958 serving as the omitted base year. The points plot year by year coefficients of *High Space Capability*$_{ij<1958}$ × *Space Industry*$_j$ interactions with the 95 percent confidence intervals indicated by the range. Space race years are 1963, 1967, and 1972. Post–space race years are 1977, 1982, 1987, and 1992. The models in all panels includes county, industry, and year fixed effects, and the count of pre-1958 patents in a county × year fixed effects. Standard errors are two-way clustered at the MSA × industry level.

involvement in private-sector patenting has many challenges, we add nonpatent controls for military involvement in local economies in columns 2, 4, 6, and 8 in panel A of Table 4. Controlling for county-level military spending or 1962 defense-funded research scientists × year fixed effects does little to alter our manufacturing point estimates or precision.

In panel B of Table 4 we add controls for worker skill. We first add controls for two measures reflecting levels of general human capital within the manufacturing sector. The fraction of nonproduction workers has the advantage that it is measured at the same unit of observation as our outcome variables: industry × county × year. It has the disadvantage, however, that it likely captures occupational as well as educational attainment variation. To capture trends that may differ by educational levels, we include the county-level high school graduate percentage in 1960 × year as controls. The results in panel B columns 1, 3, 5, and 7 show that adding these skill controls has little effect on our main space

Bayh-Dole Act was largely agency specific as Fleming et al. (2019) note. Defense funders were so-called "license agencies," which thereby enabled contractors to hold the patent title. Further, in the era we study, government interest statements were not required (see Eisenberg 1996). Thus, our military patent measures likely significantly undercount the number of military patents during the space race era. We thank a referee for making us aware of this limitation in the data.

TABLE 4—SPACE CAPABILITY AND MANUFACTURING: MILITARY AND SKILL CONTROLS

	log(value added$_{ijt}$)		log(employment$_{ijt}$)		log(capital$_{ijt}$)		log(TFP$_{ijt}$)	
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Military controls								
High space capability$_{ij<1958}$ × space race$_t$	−0.07	−0.07	−0.08	−0.08	−0.07	−0.06	0.01	0.01
	(0.07)	(0.07)	(0.06)	(0.06)	(0.09)	(0.09)	(0.03)	(0.03)
High space capability$_{ij<1958}$ × post-space race$_t$	0.02	0.03	0.05	0.05	−0.04	−0.02	−0.01	−0.01
	(0.07)	(0.07)	(0.06)	(0.06)	(0.08)	(0.08)	(0.02)	(0.02)
High space capability$_{ij<1958}$ × space race$_t$ × space industry$_j$	0.34	0.34	0.41	0.41	0.31	0.30	−0.04	−0.04
	(0.06)	(0.07)	(0.08)	(0.09)	(0.13)	(0.13)	(0.03)	(0.04)
High space capability$_{ij<1958}$ × post-space race$_t$ × space industry$_j$	0.36	0.36	0.35	0.34	0.55	0.54	−0.02	−0.02
	(0.08)	(0.08)	(0.10)	(0.10)	(0.16)	(0.18)	(0.03)	(0.04)
Additional military controls:								
Army patents$_{ijt}$	Y	Y	Y	Y	Y	Y	Y	Y
Navy patents$_{ijt}$	Y	Y	Y	Y	Y	Y	Y	Y
Military spending$_{it}$		Y		Y		Y		Y
1962 Defense scientist$_i$ × year fixed effects		Y		Y		Y		Y
R^2	0.66	0.66	0.53	0.53	0.40	0.40	0.85	0.85
Observations	6,759	6,759	6,759	6,759	6,759	6,759	6,759	6,759
Panel B. Skill controls								
High space capability$_{ij<1958}$ × space race$_t$	−0.06	−0.07	−0.07	−0.07	−0.05	−0.06	0.00	0.00
	(0.07)	(0.07)	(0.06)	(0.06)	(0.09)	(0.10)	(0.03)	(0.03)
High space capability$_{ij<1958}$ × post-space race$_t$	0.03	0.04	0.05	0.06	−0.03	−0.02	−0.01	−0.01
	(0.07)	(0.07)	(0.06)	(0.06)	(0.08)	(0.08)	(0.02)	(0.02)
High space capability$_{ij<1958}$ × space race$_t$ × space ind$_j$	0.34	0.34	0.41	0.41	0.29	0.29	−0.03	−0.03
	(0.07)	(0.09)	(0.08)	(0.10)	(0.13)	(0.13)	(0.03)	(0.04)
High space capability$_{ij<1958}$ × post-space race$_t$ × space ind$_j$	0.37	0.37	0.36	0.36	0.53	0.54	−0.02	−0.02
	(0.09)	(0.10)	(0.10)	(0.11)	(0.16)	(0.17)	(0.04)	(0.05)
Additional skill controls:								
Nonproduction worker share$_{ijt}$	Y	Y	Y	Y	Y	Y	Y	Y
1960 high school graduate$_i$ × year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
1961 IBM mainframes$_i$ × year fixed effects		Y		Y		Y		Y
1962 research scientist$_i$ × year fixed effects		Y		Y		Y		Y
R^2	0.66	0.66	0.53	0.53	0.40	0.40	0.85	0.85
Observations	6,759	6,759	6,759	6,759	6,759	6,759	6,759	6,759

Notes: See the Supplemental Data Appendix, Sections 1 and 2, for variable definitions, construction, and sources. The unit of observation is two-digit SIC industry × county × year, indexed by j, i, and t, respectively. Each column in a panel reports the results from estimating one version of equation (1) in the text. log(TFP) is defined as log(A_{ijt}) from estimating the production function $Y_{ijt} = A_{ijt}K_{ijt}^{\alpha}L_{ijt}^{\beta}$ by OLS. *High Space Capability*$_{ij<1958}$ is an indicator variable reflecting an industry-county's being above median in terms of the similarity between the technologies present in its pre-1958 patents and the CIA's intelligence estimates of Soviet space capabilities between 1958 and 1992, as described in the text and Supplemental Appendix. Space race years are 1963, 1967, and 1972. Post-space race years are 1977, 1982, 1987, and 1992. *Space Ind*$_j$ is an indicator variable for the industry j being a space industry. Main entries report coefficient estimates. Standard errors are two-way clustered at the MSA × industry level and are reported in parentheses.

race results. These variables, however, likely capture little variation in upper-tail skill that may matter for growth (Squicciarini and Voigtländer 2015). In our next set of models we add a control for the number of research scientists in a county in 1962 × year to capture differential trends in the upper-tail of human capital accumulation. We also include the number of IBM mainframes within a county in 1961 × year to capture differential trends from the installation of advanced information technology in a location. Our results remain largely unchanged across these experiments. In sum, our results on the effect of the space race on manufacturing outcomes appear highly robust to controls for local military activity and local human capital characteristics.

G. *Multipliers*

To provide some perspective on the magnitude of the effects of NASA's public R&D spending relative to government expenditures in general, we compute the contemporaneous fiscal multiplier.[32] We use the estimates in Table 3 (panel B, column 1) to compute the $\textit{Space Race Output Effect} = \widehat{\beta_4^{VA}} \times \overline{\textit{Value Added}_{ijt}} \times \overline{\textit{Output-Value Added Ratio}_{ijt}}$ and analogously a $\textit{Post-Space Race Output Effect} = \widehat{\beta_5^{VA}} \times \overline{\textit{Value Added}_{ijt}} \times \overline{\textit{Output-Value Added Ratio}_{ijt}}$. In other words, this measure computes the local value added effect associated with the highly space-capable industry-county pairs from Table 3 times the sample mean of value added but scaled up by the output/value added ratio.[33] We also compute $\textit{Space Race Spending Effect} = \widehat{\beta_4^{Spending}} \times \overline{\textit{NASA Spending}_{ijt}}$ and $\textit{Post–Space Race Spending Effect} = \widehat{\beta_5^{Spending}} \times \overline{\textit{NASA Spending}_{ijt}}$ using estimates in column 1 of Table 2, panel B. Our local fiscal multiplier estimates are then $\textit{Local Space Race Multiplier} = \frac{\textit{Space Race Output Effect}}{\textit{Space Race Spending Effect}}$ and $\textit{Local Post–Space Race Multiplier} = \frac{\textit{Post–Space Race Output Effect}}{\textit{Post–Space Race Spending Effect}}$.

We obtain an implied local fiscal multiplier for public R&D of 0.3 during the space race (i.e., 1958–1972, inclusive) and 0.4 after the space race (i.e., after 1972). Our multiplier estimates are below general government spending multipliers that Ramey (2019) concludes are in the range of 0.6–0.8. Perhaps related to space spending during the Cold War, economists have estimated multipliers in the range of 0.3–1.2 from exogenous military spending triggered by wars or foreign policy initiatives (see Batini et al. 2014). Therefore, our estimates of the effect of NASA spending are in the low range of the multiplier estimates for military spending. That our multiplier estimates tend to be comparatively small relative to the effects of other types of government spending, including defense spending, and do not increase over time indicates that the space race generated little in terms of local technological spillovers in space-related manufacturing. Moreover, the mission-oriented nature of NASA activities indicates that the R&D that contractors undertook on NASA's behalf failed to produce robust spillovers to the broader economy.

Our contemporaneous local multiplier estimates are subject to many caveats. First, our calculation does not account for the effect of NASA research and development on output in other industries or locations or how the expenditure was financed. Second, our estimates could be state dependent. The 1960s was generally a decade of economic growth, so our estimated effects could be relatively smaller than those that would have otherwise been generated in the late 1970s and 1980s when growth was slower. Third, and more broadly, our focus on NASA contractor spending does

[32] See Ramey (2021) for calibrations of long-term multiplier effects under alternative models as well as a summary of the multiplier literature with respect to public capital. Her work shows long-term multipliers are larger when the public investment has larger effects on productivity and the economy is initially below the socially optimal level of public investment. Public R&D may be expected to have a larger rate of return than other types of public spending as these conditions are more likely to be met in the public R&D case.

[33] We do not have total output in manufacturing before 1967, so we scale our value added estimates up by this fraction to find an implied total manufacturing output effect. We do not include the effects from non-space-capable industry-counties (i.e., β_2 or β_3) since these spillover effects are estimated to be zero.

not include NASA spending at universities or at NASA research centers that may have been more basic research intensive than NASA contractor spending. If technology spillovers primarily came from noncontractor NASA spending, then our approach will understate the aggregate multiplier effects of overall NASA spending. We regard our multiplier estimates, therefore, as a lower bound.

H. *Robustness*

Our results are robust to including county- and industry-specific time trends. Given the limited time dimension of the panel, however, a highly saturated model that also includes industry-county fixed effects attenuates the space race effects for valued added and employment but not capital. Adding county-year, industry-year, and industry-county fixed effects to our estimations of manufacturing outcomes is shown in Supplemental Appendix Table A8. While our definition of space industries includes transportation and electronics, we consider a broadened definition in Supplemental Appendix Table A9. Here we add printing and publishing and instruments and related products to the space industries variable. The four industries in this new measure of space industries constitute the top four space-capable industries, though only transportation and electronics were heavily demanded by NASA for the space mission. This experiment leads to overall conclusions that are largely unchanged relative to those reported above.

In our main analysis we employ two-way clustering of the standard errors at the MSA $\times$ industry level. As NASA involvement in local economies represented a localized persistent shock, we regard this clustering strategy as appropriate to address concerns of correlation in the errors term within an MSA. Within-industry clustering accounts for any unobservable correlated shocks to specific industries. Our manufacturing results are robust to inference procedures that cluster standard errors at different levels or that allow for spatial correlation in the error term (see Supplemental Appendix Table A10). The results are robust to estimation approaches that allow for heterogeneous effects (de Chaisemartin and D'Haultfoeuille 2020) that adjust standard errors to allow for correlations depending on sector-specific NASA shares (see Supplemental Appendix Table A11) or that drop a single state or industry one at a time (see Supplemental Appendix Figures A3 to A6). Further, measuring the effect on productivity in terms of growth rates rather than levels does not alter our interpretation of the results (see Supplemental Appendix Table A12).

Another way to estimate the effects of NASA contracting is to use an instrumental variables procedure. We implement this robustness check using a dummy variable for whether an industry-county received funding in year t as the endogenous variable. As shown in Supplemental Appendix Table A13, we obtain precise estimates of 4.96–6.10 on NASA contractor spending for all outcomes except productivity. We also estimate models using national industry totals by comparing NASA to non-NASA industries before and after 1958. Estimating the model at higher levels of aggregation has the strength that it includes nonlocalized effects, but the research designed is weakened—any unobserved shocks that are NASA-sector-specific could bias estimates of the impact of NASA spending. In Supplemental Appendix Table A14 we see that space-related industries were larger after the launch of Sputnik but not more productive.

Our triple difference research design utilizes changes in non-space industries as a control group where limited spillovers to non-space industries are part of the research design. We test for cross-industry spillover effects directly using a sample of only non-space industries in two ways in Supplemental Appendix Table A15. Our results show little effect of pre-1958 space technology capability after the space race began for these non-space industries, thus lending further credence to our triple difference design. We find little evidence of spillovers from space industries to their colocated non-space industry neighbors across the models.

In our main analysis we utilize the similarity of US patent documents to Soviet technology to measure the presence of space-relevant technology in an industry-county cell before 1958. Similarity to US space technology could arguably be a more relevant measure. In Supplemental Appendix Table A16 we define space capability by comparing post-Sputnik NASA patents to US space technology that existed locally prior to Sputnik. In this experiment we obtain similar estimates to those in Table 3. The dynamic effects using this US space technology-defined capability, shown in Supplemental Appendix Figure A7, again fail to reveal prior trends in manufacturing outcomes.

Moreover, we consider alternative text processing procedures and controls. We first examine the robustness of our computing the similarity of an industry-county's pre-1958 patenting to later Soviet space technology. We show in Supplemental Appendix Table A17 that our estimates are robust to how we treat terms, the rule we use to allocate a cell to treatment or control groups, and which CIA documents are included in our similarity calculation. One concern with our measure of military patents during the space race era is that government disclosure statements were not mandatory, so our measure may undercount military patents. In Supplemental Appendix Table A18 we develop a measure of patent similarity to military technology using the textual similarity of a patent to a glossary of military technology terms. Our main results are robust to these alternative ways to measure technological similarity or local military activity.

IV. Spatial Spillovers of Public R&D

Our estimated value added effects from NASA contractor spending represent the impact on the local economy rather than the national economy. To the extent we want to think about the localized space-spending infusion as a place-based policy of sorts, a question remains whether the benefits to the local economy come at a cost to other regions. Local estimates would overstate national effects if, for example, labor was supplied elastically and workers migrated toward space-related opportunities from other locations. Such an increase in employment in space locations would come at the cost of reduced employment elsewhere.[34] Such worker mobility would be consistent with historical accounts and the fact that adjustment through migration

[34] That migration can lead to different local versus national multipliers is discussed in Ramey (2019) and Chodorow-Reich (2019); however, most evidence to date has focused on less persistent spending shocks and does not find a substantial migration response. Our context may be more likely to lead to migration given the persistence of the shock to local spending from NASA's founding and continued operations as its missions evolved in the Cold War era.

can take substantial time (Blanchard and Katz 1992).[35] Alternatively, local estimates can understate national effects if there are positive demand or technology spillovers across areas.[36] How spatial spillovers may have generated a wedge between local and national effects is an empirical question.

A. *Inventor Migration*

A central challenge with measuring migration responses during the time period under consideration is lack of individual panel data.[37] We attempt to overcome these data shortcomings by using a disambiguated panel of patent inventors that tracks their locations, following the procedures in Akcigit et al. (2022). We create an individual identifier for each US inventor using patent data covering 1945–1992. See Supplemental Appendix Section 2.3 for more details. Our analysis follows Moretti and Wilson's (2017) empirical approach with three differences.[38] First, we study county-to-county migration flows within an industry and construct the data at the county × industry × patent application year level.[39] Second, our migration model includes time-invariant measures of space technology scores interacted with space era and space industry dummies. Third, we use a larger sample of inventors who are in the top 50 percent of patent producers, which enables us to employ a research design that utilizes industry variation.

Moretti and Wilson (2017) show that the equilibrium number of inventors who migrate into a county as a function of location-based factors can be estimated as:

$$
\begin{aligned}
(3)\ \log\left(\frac{P_{odjt}}{P_{oojt}}\right) = {} & \eta_1\Big(\left[\log\left(Space\ Score_{dj}\right) - \log\left(Space\ Score_{oj}\right)\right] \times Space\ Race_t\Big) \\
& + \eta_2\Big(\left[\log\left(Space\ Score_{dj}\right) - \log\left(Space\ Score_{oj}\right)\right] \\
& \quad \times Post\text{-}Space\ Race_t\Big) \\
& + \eta_3\Big(\left[\log\left(Space\ Score_{dj}\right) - \log\left(Space\ Score_{oj}\right)\right] \\
& \quad \times Space\ Race_t \times Space\ Industry_j\Big) \\
& + \eta_4\Big(\left[\log\left(Space\ Score_{dj}\right) - \log\left(Space\ Score_{oj}\right)\right] \\
& \quad \times Post\text{-}Space\ Race_t \times Space\ Industry_j\Big)
\end{aligned}
$$

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[35] For example, while almost all of the technical and clerical workers for the new Manned Spacecraft Center (renamed Johnson Space Center in 1973) in Houston could be hired locally, only 10 percent of the 6,000 scientists, engineers, and administrators were from the Houston area (Holman and Konkel 1968, pp. 31–32). Similarly, within five years of opening the center, over 125 technological firms that had a presence in the space field opened offices in Houston, including some of the most prominent such as General Electric, Honeywell, IBM, North American Aviation, Lockheed, Raytheon, Texas Instruments, and TRW (Brady 2007, p. 455).

[36] Myers and Lanahan (2022) find positive technological spillovers across space, and positive demand spillovers are at the heart of the market access approach developed in Donaldson and Hornbeck (2016).

[37] The 1940s to 1960s is too recent for linked population census data to be available and too early for modern panel datasets, such as the Panel Study of Income Dynamics (PSID), that track an individual's location.

[38] We choose to follow Moretti and Wilson (2017) instead of Akcigit et al. (2022) as the latter's approach has a significant computational burden at the state level and we are using even more fine-grained county-level data.

[39] In this context patent application year is preferred over patent grant year that we use above as it is closer to the time period of innovation. We thus obtain a measure of location with less measurement error by using application year instead of grant year. We use the modal industry across all patents filed by an inventor to classify them by sector.

$$+ \eta_5\left[\log(1 - I_{dt}) - \log(1 - I_{ot})\right]$$

$$+ \eta_6\left[\log(1 - C_{dt}) - \log(1 - C_{ot})\right]$$

$$+ \eta_7\left[\log(1 + R_{dt}) - \log(1 + R_{ot})\right] + \textit{Pre-1958}\ \textit{Patents}_{oj} \times \gamma_t$$

$$+ \textit{Pre-1958}\ \textit{Patents}_{dj} \times \gamma_t + \gamma_{od} + \gamma_o + \gamma_d + \gamma_t + \gamma_j + \mu_{odjt}.$$

We denote origin locations o and destination locations d. The number of inventors who move from o to d in industry j is P_{odjt}, and the number of inventors in industry j who begin in o and do not move is P_{oojt}, so that $\log\left(P_{odjt}/P_{oojt}\right)$ is the log odds ratio for inventor out-migration. We examine how the odds of moving depend on the differences in space scores, $\left[\log(\textit{Space Score}_{dj}) - \log(\textit{Space Score}_{oj})\right]$, interacted with indicator variables for the space race and post–space race periods. We control for origin-destination differentials in personal income tax rates, $\left[\log(1 - I_{dt}) - \log(1 - I_{ot})\right]$, corporate income tax rates, $\left[\log(1 - C_{dt}) - \log(1 - C_{ot})\right]$, and R&D tax credits, $\left[\log(1 + R_{dt}) - \log(1 + R_{ot})\right]$. Finally, we control for county origin (γ_o) and destination (γ_d) fixed effects, year of patent application (γ_t) fixed effects, industry fixed effects (γ_j), and pair fixed effects (γ_{od}) to capture time-invariant pair-specific features such as distance or travel costs.[40] To account for trends by initial innovation intensity, as in our analysis above, we also control for both origin and destination pre-1958 patent count in the industry-county times year fixed effects ($\textit{Pre-1958}\ \textit{Patents}_{oj} \times \gamma_t$ and $\textit{Pre-1958}\ \textit{Patents}_{dj} \times \gamma_t$, respectively). We consider a few variants of this specification—with and without tax rates and including state × year fixed effects—in our analysis.

The coefficient estimates η_3 and η_4 capture how the relationship between space capability score differentials between origin and destination industry-county pairs affected migration during and after the space race relative to the pre-NASA era. If NASA spending caused inventors to migrate toward space-capable places, then we would expect η_3 and η_4 to be positive. Time invariant factors that affected wages or amenities in the origin and destination locations, as well as typical migration patterns, are controlled using origin (γ_o), destination (γ_d), and pairwise (γ_{od}) fixed effects. A potential threat to our approach would occur if changes in wages or amenities during and after the space race were correlated with differentials in ex ante space capabilities. Based on our results above and historical accounts, we do not expect this issue to be likely.

Table 5 reports the results of estimating alternative versions of our migration model. In column 1 we see that NASA spending caused inventors to move toward areas with relatively higher space capability scores in the space race and post-race periods. That the post–space race effects are larger may indicate it takes some time for researchers to adjust to a demand shock through migration. Adding controls for personal tax rates, corporate tax rates, and R&D tax credits in column 2 does little

[40] For this analysis, we follow Moretti and Wilson (2017) in showing standard errors that allow for three-way clustering by origin county × year, destination county × year, and origin-destination pair. This clustering addresses the issues that errors could be correlated across origin (destination) counties within a year because they share the same level of space technology similarity in all observations involving that origin (destination) county in a year. In addition, standard errors may be correlated over time within the panel.

TABLE 5—SPACE CAPABILITY DIFFERENCES AND PATENT INVENTOR MIGRATION

	log(out migration ratio$_{odjt}$)		
Dependent variable	(1)	(2)	(3)
Space capability score difference$_{odj,<1958}$ × space race$_t$	−0.15	−0.13	−0.14
	(0.05)	(0.05)	(0.06)
Space capability score difference$_{odj,<1958}$ × post-space race$_t$	−0.18	−0.15	−0.16
	(0.12)	(0.12)	(0.11)
Space capability score difference$_{odj,<1958}$ × space race$_t$ × space industry$_j$	0.27	0.27	0.28
	(0.10)	(0.10)	(0.10)
Space capability score difference$_{odj,<1958}$ × post–space race$_t$ × space industry$_j$	0.68	0.62	0.52
	(0.23)	(0.23)	(0.24)
Corporate income tax rate $(1 - \text{CIT})_{odt}$		−2.15	
		(0.84)	
Personal average income tax rate, 90th percentile $(1 - \text{ATR})_{odt}$		1.72	
		(0.53)	
R&D credit $(1 + \text{credit})_{odt}$		−0.04	
		(0.06)	
Origin county fixed effects	Y	Y	Y
Destination county fixed effects	Y	Y	Y
Industry fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Origin pre-1958 patents × year fixed effects	Y	Y	Y
Destination pre-1958 patents × year fixed effects	Y	Y	Y
Origin county × destination county fixed effects	Y	Y	Y
Origin state × year fixed effects			Y
Destination state × year fixed effects			Y
R^2	0.75	0.75	0.80
Observations	11,950	11,950	11,950

Notes: See the Supplemental Data Appendix, Sections 1 and 2, for variable definitions, construction, and sources. The unit of observation is origin county × destination county × industry × application year. Each column in the table reports the results from estimating one version of equation (3) in the text. *Space Capability Score Difference*$_{odj<1958}$ = log(*Space Capability Score*$_{dj<1958}$) − log(*Space Capability Score*$_{oj<1958}$) is the difference in space capability scores between the origin and destination counties in industry j, as described in the text and Supplemental Appendix. Space race years are 1963, 1967, and 1972. Post–space race years are 1977, 1982, 1987, and 1992. Standard errors in parentheses, with three-way clustering by origin county × year, destination county × year, and county-pair.

to alter these results. Finally, column 3 adds origin state × application year and destination state × application year fixed effects. Across all of these specifications our results change little and the robust conclusion is that space race spending in space industries led to inventors' migration toward opportunity, which is consistent with the employment effects found in Table 3 and with historical accounts. It is interesting to note the negative and statistically significant effect of the main space capability variable during the space race era. Non-space sector inventors may have faced competition for scarce laboratory resources or opportunities in the short run. Understanding these indirect effects that local inventors might have faced is an interesting topic for future research.

B. *Including Market Effects*

How might migration, demand, and technology spillovers combine to affect the national return to R&D spending? To address this question we incorporate

market-level effects of R&D that might generate a wedge between local and national effects driven by R&D spending in other counties. These market-level effects are derived in an extension to the simple county-to-county trade model from Donaldson and Hornbeck (2016) in Supplemental Appendix Section 4.[41] The theoretical framework leads to the following estimating equation:

$$
\begin{aligned}
(4)\quad Y_{ijt} = {} & \beta_1 + \beta_2 \textit{High Space Capability}_{ij<1958} \times \textit{Space Race}_t \\
& + \beta_3 \textit{High Space Capability}_{ij<1958} \times \textit{Post-Space Race}_t \\
& + \beta_4 \textit{High Space Capability}_{ij<1958} \times \textit{Space Race}_t \times \textit{Space Industry}_j \\
& + \beta_5 \textit{High Space Capability}_{ij<1958} \times \textit{Post-Space Race}_t \\
& \times \textit{Space Industry}_j + \beta_6 \textit{High Space Market}_{ij<1958} \times \textit{Space Race}_t \\
& + \beta_7 \textit{High Space Market}_{ij<1958} \times \textit{Post-Space Race}_t \\
& + \beta_8 \textit{High Space Market}_{ij<1958} \times \textit{Space Race}_t \times \textit{Space Industry}_j \\
& + \beta_9 \textit{High Space Market}_{ij<1958} \times \textit{Post-Space Race}_t \times \textit{Space Industry}_j \\
& + \textit{Total Pre-1958 Patents}_{ij} \times \gamma_t + \delta_i + \theta_j + \gamma_t + \upsilon_{ijt}.
\end{aligned}
$$

We define $\textit{High Space Market}_{ij<1958}$ as a binary variable where county-industries with above median values of our space-score-based market measure receive a value of one and other counties receive a zero. For details of how this variable is constructed see Supplemental Appendix Section 4.2. Our goal is to estimate β_8 and β_9, which will capture the market-level effects of space race activity elsewhere during and after the race to the moon that may have affected space industries locally. With these estimates in hand we can get a sense of how spatial spillovers may affect our estimates of the fiscal multiplier reported above.

In Table 6 we report the results of estimating equation (4). The results show that including controls for market-level effects does little to alter the local space-capability effects estimated above. Their magnitudes are little changed and remain precisely estimated. The point estimates for the market effects are quite close to zero, with signs that are outcome or specification dependent and imprecisely estimated. A lack of market effects would be consistent with the worker mobility toward space county-industries described above, which seem to have counterbalanced any positive market-level demand or technology spillover effects. These results indicate that the lack of spatial spillovers on net implies that the local impact of NASA R&D

[41] This approach allows us to quantify national effects while maintaining research design credibility typically found in reduced-form studies. We differ from Donaldson and Hornbeck (2016), however, in that we focus on the impact of public R&D spending, holding transportation infrastructure fixed and introducing market-level consumption externalities.

TABLE 6—SPACE CAPABILITY AND MANUFACTURING: LOCAL AND MARKET EFFECTS

	log(value added$_{ijt}$)		log(employment$_{ijt}$)		log(capital$_{ijt}$)		log(TFP$_{ijt}$)	
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
High space capability$_{ij<1958}$ × space race$_t$	−0.07	−0.08	−0.08	−0.09	−0.06	−0.03	0.01	0.00
	(0.07)	(0.07)	(0.06)	(0.07)	(0.09)	(0.09)	(0.03)	(0.02)
High space capability$_{ij<1958}$ × post-space race$_t$	0.02	0.02	0.04	0.04	−0.05	0.00	0.00	−0.01
	(0.07)	(0.08)	(0.06)	(0.07)	(0.08)	(0.09)	(0.02)	(0.02)
High space capability$_{ij<1958}$ × space race$_t$ × space industry$_j$	0.32	0.38	0.39	0.47	0.28	0.26	−0.04	−0.03
	(0.07)	(0.09)	(0.10)	(0.11)	(0.12)	(0.15)	(0.04)	(0.05)
High space capability$_{ij<1958}$ × post-space race$_t$ × space industry$_j$	0.49	0.42	0.50	0.44	0.63	0.59	−0.04	−0.05
	(0.12)	(0.12)	(0.12)	(0.14)	(0.16)	(0.17)	(0.04)	(0.04)
High space market$_{ij<1958}$ × space race$_t$	0.04	0.04	0.07	0.07	−0.02	−0.04	−0.01	−0.01
	(0.09)	(0.10)	(0.09)	(0.10)	(0.17)	(0.16)	(0.03)	(0.03)
High space market$_{ij<1958}$ × post-space race$_t$	0.12	0.12	0.19	0.21	0.16	0.14	−0.06	−0.07
	(0.13)	(0.13)	(0.11)	(0.11)	(0.20)	(0.19)	(0.04)	(0.05)
High space market$_{ij<1958}$ × space race$_t$ × space industry$_j$	0.05	0.00	0.08	0.01	0.00	−0.12	−0.01	0.02
	(0.06)	(0.07)	(0.06)	(0.07)	(0.07)	(0.06)	(0.03)	(0.04)
High space market$_{ij<1958}$ × post-space race$_t$ × space industry$_j$	−0.11	0.00	−0.10	0.03	0.04	−0.05	−0.05	−0.01
	(0.09)	(0.10)	(0.09)	(0.09)	(0.07)	(0.11)	(0.04)	(0.05)
County fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Pre-1958 patents$_i$ × year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
MSA$_i$ × year fixed effects		Y		Y		Y		Y
R^2	0.66	0.68	0.53	0.56	0.40	0.46	0.85	0.86
Observations	6,759	6,759	6,759	6,759	6,759	6,759	6,759	6,759

Notes: See the Supplemental Data Appendix, Sections 1 and 2, for variable definitions, construction, and sources. The unit of observation is two-digit SIC industry × county × year, indexed by *j*, *i*, and *t*, respectively. Each column in the table reports the results from estimating one version of equation (4) in the text. log(TFP) is defined as $\log(A_{ijt})$ from estimating the production function $Y_{ijt} = A_{ijt}K_{ijt}^{\alpha}L_{ijt}^{\beta}$ by OLS. *High Space Capability*$_{ij<1958}$ is an indicator variable reflecting an industry-county's being above median in terms of the similarity between the technologies in pre-1958 patents and the intelligence estimates of Soviet space capabilities between 1958 and 1992, as described in the text and Supplemental Appendix. *High Space Market*$_{ij<1958}$ takes a value of one in industry-counties with above median space capability score in their market, as described in Section 1.2 of the Supplemental Appendix. Space race years are 1963, 1967, and 1972. Post–space race years are 1977, 1982, 1987, and 1992. *Space Industry*$_j$ is an indicator variable for the industry *j* being a space industry. Main entries report coefficient estimates. Standard errors are two-way clustered at the MSA × industry level and are reported in parentheses.

spending that we estimated above is a reasonable proxy for NASA's impact on the broader economy.

V. Conclusion

Landing on the moon in 1969 represented a critical moment for boosting American technological capabilities and leadership. Looking to this iconic moonshot event, our paper seeks to address fundamental questions about the role of public R&D in facilitating economic growth, both locally and more broadly. Despite its focal point as a shining example of American R&D investment and accomplishment, there is no credible empirical estimate of the space mission's contribution to economic growth. Using newly collected data and a novel identification strategy that takes advantage of the geopolitical tensions of the historic moment, we uncover economically meaningful, stable, and precisely estimated effects of public R&D on long-term manufacturing growth in the space sector. Yet the magnitudes of the estimated effects are fairly modest and seem no larger than the impacts from other non-R&D types of government expenditures.

While we show significant, though modest, positive effects from NASA contractor spending during and after the moonshot era, some caution is warranted in applying our estimates to public R&D more broadly. As Mowery, Nelson, and Martin (2010) note, mission-oriented R&D is unusually focused on a specific goal and highly centralized. Whether non-mission-oriented public R&D would generate similar returns to those of NASA's space race remains an open question. Similarly, our focus on NASA contractor spending enables a tight research design and captures the majority of NASA spending but at the same time may be limited in capturing all of NASA's technological spillovers. Public R&D spending at NASA centers or in universities may have been more for basic science and a more important source of enduring technological spillovers that our approach does not capture.

Economists have long sought to untangle the multiple factors that contribute to economic growth. The roles of public and private sector R&D, human and physical capital investment, transportation and communications infrastructure, culture, geography, political and legal institutions, and even luck have been carefully explored and debated. Our analysis of the space race and its aftermath sheds some light on the limits of public R&D in generating economic growth. Today the US government invests a tiny fraction in non-military R&D relative to the heights of the Cold War. To the extent that some policymakers and advisors call for a new Sputnik Moment to seed a new era of US economic growth in targeted sectors, our analysis of the economic impacts of the politically-charged space race era investments offers important cautionary lessons.

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